Pacific Herring preliminary data summary for Strait of Georgia 2026

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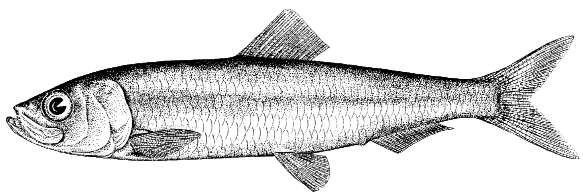

Pacific Herring (*Clupea pallasii*). Image credit: Fisheries and Oceans Canada.

Disclaimer This report contains preliminary data collected for Pacific Herring in 2026 in the Strait of Georgia major stock assessment region (SAR). These data may differ from data used and presented in the final stock assessment.

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1 Context

Pacific Herring (*Clupea pallasii*) in British Columbia are assessed as 5 major and 2 minor stock assessment regions (SARs), and data are collected and summarized on this scale (Table 1, Figure 1). Note that formal stock assessments are only done for major SARs. The Pacific Herring data collection program includes fishery-dependent and -independent data from 1951 to 2026. This includes annual time series of commercial catch data, biological samples (providing information on proportion-at-age and weight-at-age), and spawn index data conducted using a combination of surface and SCUBA surveys. In some areas, industry- and/or First Nations-operated in-season soundings programs are also conducted, and this information is used by resource managers, First Nations, and

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stakeholders to locate fish and identify areas of high and low Pacific Herring biomass to plan harvesting activities. In-season acoustic soundings are not used by stock assessment to inform the estimation of spawning biomass.

The following is a description of data collected for Pacific Herring in 2026 in the Strait of Georgia major SAR (Figure 2). Data collected outside the SAR boundary are not included in this summary, and are not used for the purposes of stock assessment. Although we summarise data at the scale of the SAR for stock assessments, we summarise data at finer spatial scales in this report: Locations are nested within Sections, Sections are nested within Statistical Areas, and Statistical Areas are nested within SARs (Table 2). For the Strait of Georgia major SAR, we use another level of spatial aggregation which we refer to as a 'Group'. Note that we refer to 'year' instead of 'herring season' in this report; therefore 2026 refers to the 2025/2026 Pacific Herring season.

2 Data collection programs

Biological samples were collected by the seine charter vessel *Denman Isle* for 25 days from February 20th to March 16th. Four additional industry test vessels collected biological samples from March 1st to March 14th: *Viking Cavalier*, *Nita Maria*, *Western Investor*, and *Alaska Queen*. The primary purpose of the test charter vessel was to collect biological samples from main aggregations of herring in the SoG, as identified from soundings within the SAR boundary.

Herring spawn locations were primarily identified with fixed-wing overflights conducted by DFO Resource Management and Science staff. Thirteen flights were conducted this season, all in March.

Three dive vessels operated in the SoG:

- The *Pacific Discovery* surveyed 18 days from March 13th to March 30th,
- The *Ocean Cloud* surveyed 12 days from March 12th to March 23rd, and
- Divers with A-Tlegay Fisheries Society surveyed 1 day on March 9th.

The first two dive vessels and the seine charter vessel *Denman Isle* were funded by DFO, through a contract to the Herring Conservation Research Society. Additional sampling and sounding efforts conducted through the industry test program were contributed in-kind by the herring industry.

3 Catch and biological samples

In the 1950s and 1960s, the reduction fishery dominated Pacific Herring catch; starting in the 1970s, catch has been predominantly from roe seine and gillnet fisheries. The reduction fishery is different from current fisheries in several ways. First, the reduction fishery caught Pacific Herring of all ages, whereas current fisheries target spawning (i.e., mature) fish. Thus, reduction fisheries included an unknown number of age-1 fish which are not typically caught in current fisheries. Second, the reduction fishery has some uncertainty regarding the quantity and location of catch; however catches have been resolved to SAR and Statistical Area using fish slips as best as possible. For the roe gillnet fishery, all Pacific Herring catch has been validated by a dockside monitoring program since 1998; the catch validation program started in 1999 for the roe seine fishery. Finally, the reduction fishery operated during the summer and winter months, whereas roe fisheries typically target spawning fish between February and April.

Landed commercial catch of Pacific Herring by year and fishery is shown in Table 3 and Figure 3. Total harvested spawn-on-kelp (SOK) in 2026 in the Strait of Georgia major SAR is shown in Table 4; we also calculate the estimated spawning biomass associated with SOK harvest. See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. Incidental mortality from fisheries and aquaculture activities is shown in Figure 4.

In 2026, 127 Pacific Herring biological samples were collected and processed for the Strait of Georgia major SAR (Table 5, Table 6). The locations in which the biological samples were collected are presented in Figure 5. Figure 6 shows the distribution of fish weight by year and sample type. Included herein are biological summaries of observed proportion-, number-, weight-, and length-at-age (Figure 7, Table 7, and Figure 8, respectively). We also show the percent change in weight and length for age-3 and age-6 fish (Figure 9 & Figure 10, respectively). Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. Only representative biological samples are included, where 'representative' indicates whether the Pacific Herring sample in the set accurately reflects the larger Pacific Herring school.

3.1 Nearshore genetics project

DFO initiated a nearshore genetics project in 2022 to better understand the spatial and temporal structure of Pacific Herring stocks in British Columbia. We do this by collecting genetic and biological data from actively spawning Pacific Herring (*Clupea pallasii*). Then we collect biological data (e.g., length, weight, age, sex) and genetic data from the fish. Finally, we analyse the biological and genetic data to better understand Pacific Herring. Genetic samples were collected in the Strait of Georgia major SAR in 2023 and 2024 (Figure 11). Results from the nearshore genetics project will be presented in a separate document when they are available.

4 Spawn survey data

Pacific Herring spawn surveys were conducted at 48 individual locations in 2026 in the Strait of Georgia major SAR (Table 8, and Figure 12). A summary of spawn from the last decade (2016 to 2025) is shown in Figure 13. Figure 14 shows spawn start date by decade and Group. Spawn surveys are conducted to estimate the spawn length, width, number of egg layers, and substrate type, and these data are used to estimate the index of spawning biomass (i.e., the spawn index; Figure 15, Figure 16, Figure 17, Figure 18,

Table 9, and Figure 19). See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Therefore, these data do not represent model estimates of spawning biomass, and are considered the minimum observed spawning biomass derived from egg counts. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

Some Pacific Herring Sections contribute more than others to the total spawn index, and the percentage contributed by Section varies yearly (Figure 18b, Figure 20). For example, in 2026, Section 172 contributed the most to the spawn index (36%). As with Sections, some Groups contribute more than others to the total spawn index (Figure 18c, Figure 21). An animation shows the spawn index by spawn survey location from 1951 to 2026 (Figure 22).

5 First Nations observations

The following observations were contributed by representatives of First Nations communities. These observations provide context and perspectives to this data report. In some cases we make edits for clarity and brevity, but we do not change the intent or substance of responses. Note: if your observations are missing, submit them here.

5.1 Qualicum

Surveyed for spawn from March 6th to 13th from Campbell River down to Icarus Point. Caught biosamples in heavy active spawn off Gurney Point, in heavy spawn off Collishaw Point, off Phipps Point, Icarus Point, and Blunded Point. Collected water samples inside Baynes Sound but not fish observed, at Nile Creek and sounded fish on the bottom, and off Blunden Point.

Sounded some small schools in Kye Bay and Deep Bay, and found washed out spawn off Chrome Island, on the West side of Flora Island, and heavy spawn off Whaling Station Bay. Sounded small schools of spawned out fish off Ferry Landing on Hornby. Found scattered fish on the bottom between Nile Creek and Bowser, and three schools off Qualicum Beach and one at the mouth of Northwest Bay. Found three schools between Mistaken Island and French Creek harbour. Medium spawns off Nanoose Bay entrance and Maude Island.

5.2 Tla'amin Nation

The Guardians prepped for setting boughs and harvest, however missed the opportunity to set during the spawns around Savary. The Guardians did practice and teach prepping the boughs and setting with the GIJE highschool program for Indigenous youth.

6 General observations

The following observations were reported by DFO Resource Management staff and DFO Science staff. These observations provide additional context to this data report.

- Temporal distribution of spawn from Feb 26th to Mar 22nd.
- Similar to last year, fish were not found in large aggregations by sounding platforms prior to spawning.
- Divers with A-Tlegay Fisheries Society surveyed spawn on Savary Island.
- Spawn observed south of Dodd Narrows.

7 Tables

Table 1. Pacific Herring stock assessment regions (SARs) in British Columbia.

Name	Code	Type
Haida Gwaii	HG	Major
Prince Rupert District	PRD	Major
Central Coast	CC	Major
Strait of Georgia	SoG	Major
West Coast of Vancouver Island	WCVI	Major
Area 27	A27	Minor
Area 2 West	A2W	Minor

Table 2. Statistical Areas, Sections, and Groups for Pacific Herring in the Strait of Georgia major stock assessment region (SAR). Legend: '14&17' is Statistical Areas 14 and 17 (excluding Section 173); 'ESoG' is eastern Strait of Georgia; 'Lazo' is above Cape Lazo; and 'SDodd' is South of Dodd Narrows

Region	Statistical Area	Section	Group
Strait of Georgia	13	132	Lazo
Strait of Georgia	13	135	Lazo
Strait of Georgia	14	140	14&17
Strait of Georgia	14	141	Lazo
Strait of Georgia	14	142	14&17
Strait of Georgia	14	143	14&17
Strait of Georgia	15	150	ESoG
Strait of Georgia	15	151	ESoG
Strait of Georgia	15	152	ESoG
Strait of Georgia	16	160	ESoG
Strait of Georgia	16	161	ESoG
Strait of Georgia	16	162	ESoG
Strait of Georgia	16	163	ESoG
Strait of Georgia	16	164	ESoG
Strait of Georgia	16	165	ESoG
Strait of Georgia	17	170	14&17
Strait of Georgia	17	171	14&17
Strait of Georgia	17	172	14&17
Strait of Georgia	17	173	Sec173
Strait of Georgia	18	180	SDodd
Strait of Georgia	18	181	SDodd
Strait of Georgia	18	182	SDodd

Table 2 continued

Region	Statistical Area	Section	Group
Strait of Georgia	19	190	SDodd
Strait of Georgia	19	191	SDodd
Strait of Georgia	19	192	SDodd
Strait of Georgia	19	193	SDodd
Strait of Georgia	28	280	ESoG
Strait of Georgia	29	291	ESoG
Strait of Georgia	29	292	ESoG

Table 3. Total landed commercial catch of Pacific Herring in metric tonnes (t) by gear type in 2026 in the Strait of Georgia major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Note: data may be withheld due to privacy concerns (WP).

Gear	Catch (t)
Other	2,168
RoeSN	5,563
RoeGN	5,043

Table 4. Total harvested Pacific Herring spawn-on-kelp (SOK) in pounds (lb), and the associated estimate of spawning biomass in metric tonnes (t) from 2016 to 2026 in the Strait of Georgia major stock assessment region (SAR). See the spawn index technical report for calculations to convert SOK harvest to spawning biomass. Note: data may be withheld due to privacy concerns (WP).

Year	Harvest (lb)	Spawning biomass (t)
2016	0	0
2017	0	0
2018	0	0
2019	0	0
2020	0	0
2021	0	0
2022	0	0
2023	0	0
2024	0	0
2025	0	0
2026	0	0

Table 5. Number of Pacific Herring biological samples processed from 2016 to 2026 in the Strait of Georgia major stock assessment region (SAR). Each sample is approximately 100 fish. Note: Nearshore samples are not used in stock assessments.

	Number of samples			
Year	Commercial	Test	Nearshore	Total
2016	123	33	5	161
2017	121	27	0	148
2018	116	28	0	144
2019	125	35	0	160
2020	106	27	0	133
2021	89	25	0	114
2022	50	18	2	70
2023	73	21	22	116
2024	68	24	16	108
2025	66	20	4	90
2026	111	16	0	127

Table 6. Number and type of Pacific Herring biological samples processed in 2026 in the Strait of Georgia major stock assessment region (SAR). Each sample is approximately 100 fish.

Type	Gear	Use	Number of samples
Commercial	Gillnet	Roe fishery	55
Commercial	Seine	Bait fishery	2
Commercial	Seine	Food fishery	17
Commercial	Seine	Roe fishery	37
Test	Seine	Test fishery	16

Table 7. Observed proportion-at-age for Pacific Herring from 2016 to 2026 in the Strait of Georgia major stock assessment region (SAR). The age-10 class is a 'plus group' which includes fish ages 10 and older.

	Proportion-at-age								
Year	2	3	4	5	6	7	8	9	10
2016	0.153	0.267	0.334	0.178	0.040	0.016	0.009	0.001	0.001
2017	0.100	0.304	0.267	0.215	0.089	0.018	0.006	0.002	0.000
2018	0.077	0.418	0.247	0.144	0.082	0.026	0.004	0.002	0.000
2019	0.238	0.349	0.222	0.110	0.051	0.023	0.006	0.001	0.000
2020	0.124	0.529	0.208	0.081	0.035	0.016	0.004	0.002	0.000
2021	0.052	0.372	0.391	0.124	0.041	0.013	0.005	0.001	0.001
2022	0.240	0.282	0.238	0.176	0.048	0.010	0.004	0.001	0.001
2023	0.106	0.390	0.227	0.161	0.097	0.013	0.005	0.000	0.000
2024	0.079	0.375	0.327	0.114	0.071	0.027	0.005	0.002	0.000
2025	0.064	0.372	0.317	0.163	0.057	0.017	0.009	0.001	0.000
2026	0.069	0.357	0.303	0.161	0.076	0.025	0.006	0.002	0.000

Table 8. Pacific Herring spawn survey locations, start date, and spawn index in metric tonnes (t) in 2026 in the Strait of Georgia major stock assessment region (SAR). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (NAs).

Statistical Area	Section	Location name	Start date	Spawn index (t)
14	141	Kitty Coleman Beach	March 03	5,883
14	141	Kye Bay	March 03	4,624
14	141	Little Rvr	March 05	872
14	142	Cape Lazo	March 05	4,700
14	142	Collishaw Pt	March 05	13,935
14	142	Fillongley Park	March 20	214
14	142	Gravelly Bay	March 18	556
14	142	Shingle Spit	March 07	608
14	142	Tralee Pt	March 05	2,689
14	142	Whalebone Pt	March 18	889
14	142	Whaling Station Bay	March 03	1,743
14	143	Columbia Beach	March 19	3,434
14	143	French Cr	March 19	2,134
14	143	Parksville Bay	March 20	158
14	143	Qualicum Beach	March 19	1,480
15	152	Hernando Is	March 05	NA
15	152	Savary Is	February 26	633
16	163	Pender Hrbr	January 25	NA
16	165	Four Mile Pt	February 01	NA
17	172	Blunden Pt	March 12	1,970
17	172	Gabriola Is (N Shore)	March 29	NA
17	172	Hammond Bay	March 17	961
17	172	Horswell Bluff	March 17	800
17	172	Icarus Pt	March 13	7,686
17	172	Lagoon Hd	March 17	969
17	172	Lantzville	March 12	1,086
17	172	Maude Is	March 12	1,063
17	172	Nankivell Pt	March 12	7
17	172	Neck Pt	March 15	1,133
17	172	Ranch Pt	March 09	176
17	172	Ruth Is	March 12	549
17	172	Schooner Cv	March 12	2,268
17	172	Southey Is	March 12	1,510
17	172	Sunrise Beach	March 13	11,744
17	172	Wallis Pt	March 08	3,519
17	173	Clam Bay	March 20	447
17	173	Danger Rfs	March 20	1,757
17	173	Fraser Pt	March 18	7,403

Table 8 continued

Statistical Area	Section	Location name	Start date	Spawn index (t)
17	173	Kulleet Bay	March 20	203
17	173	Miami Islets	March 18	1,662
17	173	North Cv	March 18	1,951
17	173	Pilkey Pt	March 18	1,883
17	173	Preedy Hrbr	March 20	2,421
17	173	Ragged Islets	March 18	296
17	173	Telegraph Hrbr	March 20	296
17	173	Yellow Pt	March 20	754
18	181	Burgoyne Bay	March 03	NA
19	193	Esquimalt Hrbr	March 30	NA

Table 9. Summary of Pacific Herring spawn survey data from 2016 to 2026 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Units: metres (m), and metric tonnes (t).

Year	Total length (m)	Mean width (m)	Mean number of egg layers	Spawn index (t)
2016	118,300	157	1.2	129,501
2017	130,440	162	0.9	81,063
2018	87,745	168	1.4	91,938
2019	82,865	139	1.1	63,037
2020	154,345	133	1.4	116,150
2021	114,950	154	1.1	70,938
2022	78,995	168	1.4	86,114
2023	102,550	142	1.4	74,506
2024	123,325	129	1.4	81,774
2025	138,375	130	1.6	97,903
2026	100,650	131	2.0	99,088

8 Figures

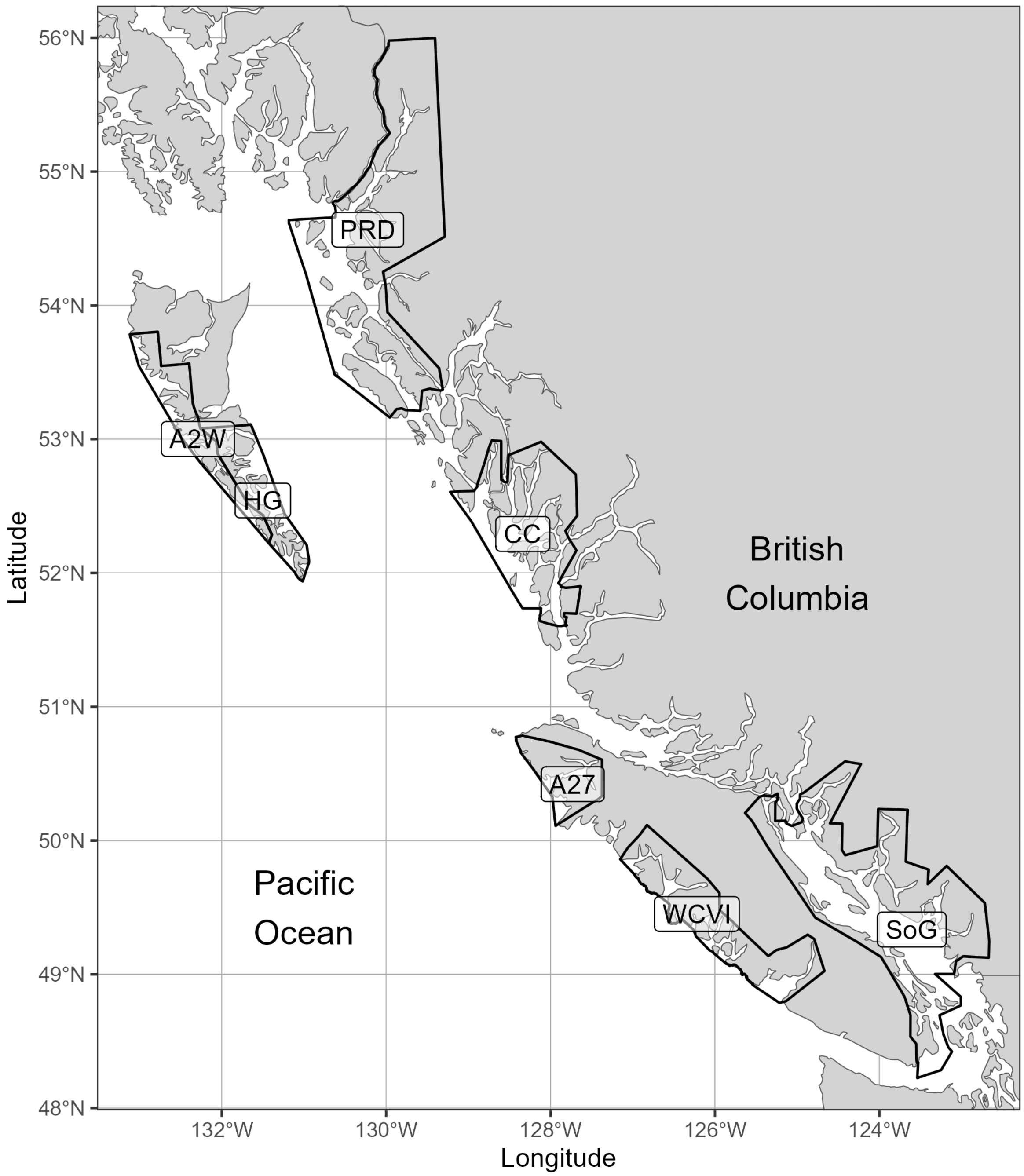

Figure 1. Boundaries for the Pacific Herring stock assessment regions (SARs) in British Columbia. There are 5 major SARs: Haida Gwaii (HG), Prince Rupert District (PRD), Central Coast (CC), Strait of Georgia (SoG), and West Coast of Vancouver Island (WCVI). There are 2 minor SARs: Area 27 (A27) and Area 2 West (A2W). Units: kilometres (km).

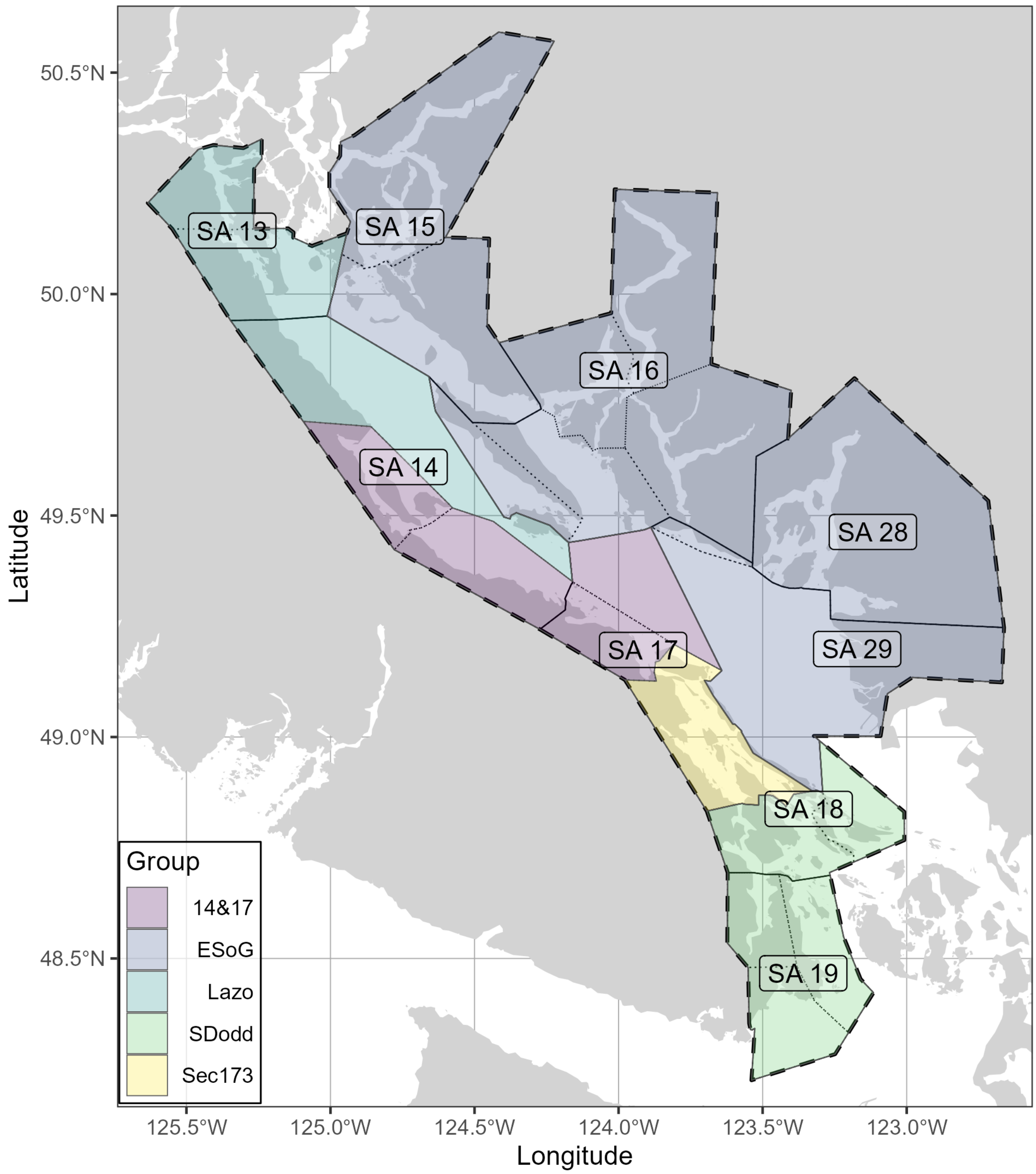

Figure 2. Boundaries for the Strait of Georgia major stock assessment region (SAR; thick dashed lines), associated Statistical Areas (SA; thin solid lines), and associated Sections (thin dotted lines). Units: kilometres (km). Legend: '14&17' is Statistical Areas 14 and 17 (excluding Section 173); 'ESoG' is eastern Strait of Georgia; 'Lazo' is above Cape Lazo; and 'SDodd' is South of Dodd Narrows

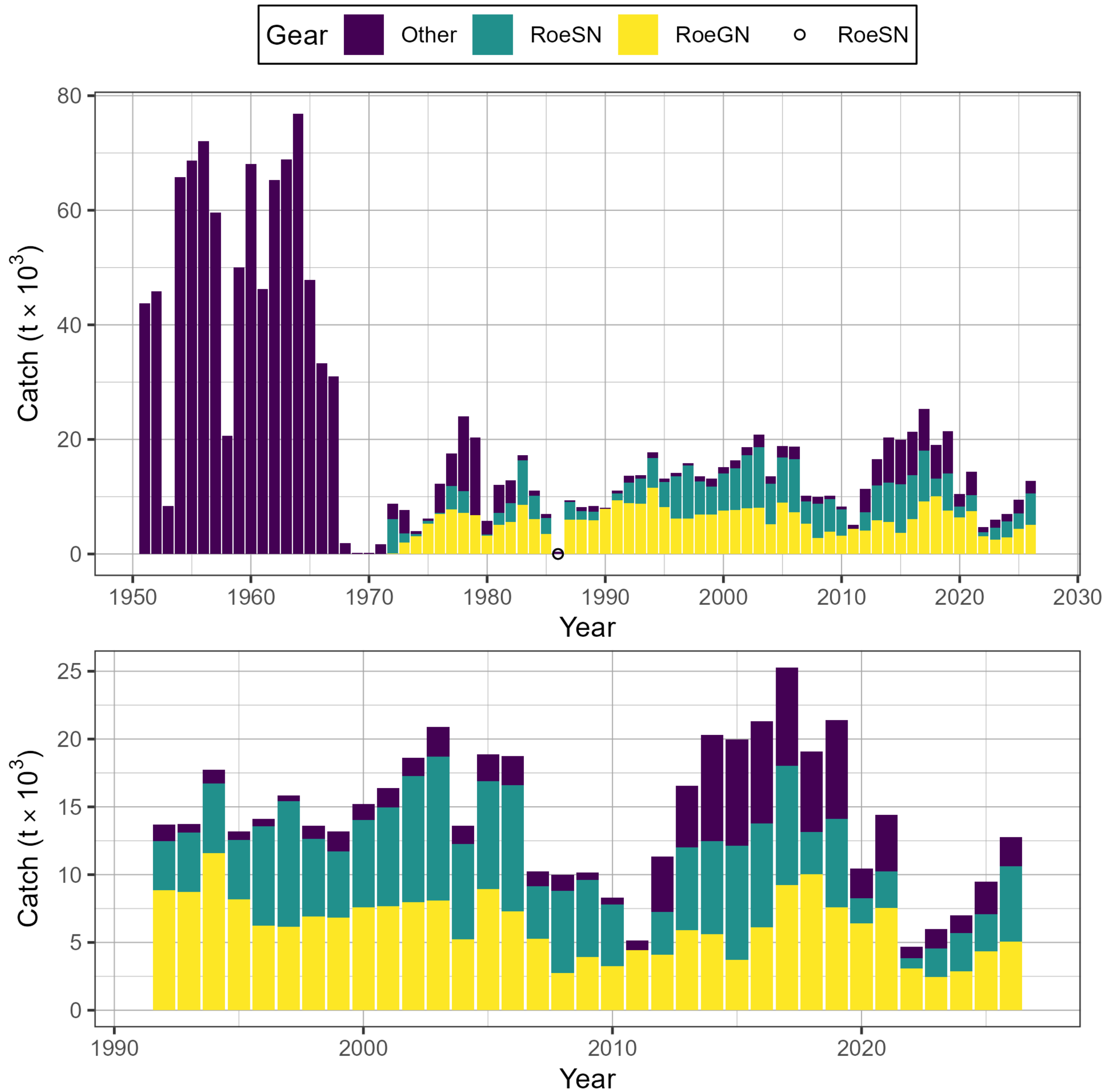

Figure 3. Time series of total landed catch in thousands of metric tonnes (t $\times$ 10^3) of Pacific Herring by gear type from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). Legend: 'Other' represents the reduction (1951 to 1970 only), the food and bait, as well as the special use fishery; 'RoeSN' represents the roe seine fishery; and 'RoeGN' represents the roe gillnet fishery. Data from the spawn-on-kelp (SOK) fishery are not included. Bottom panel shows catch since 1991 in more detail. Note: symbols indicate years in which catch by gear type (i.e., Other, RoeSN, RoeGN) is withheld due to privacy concerns.

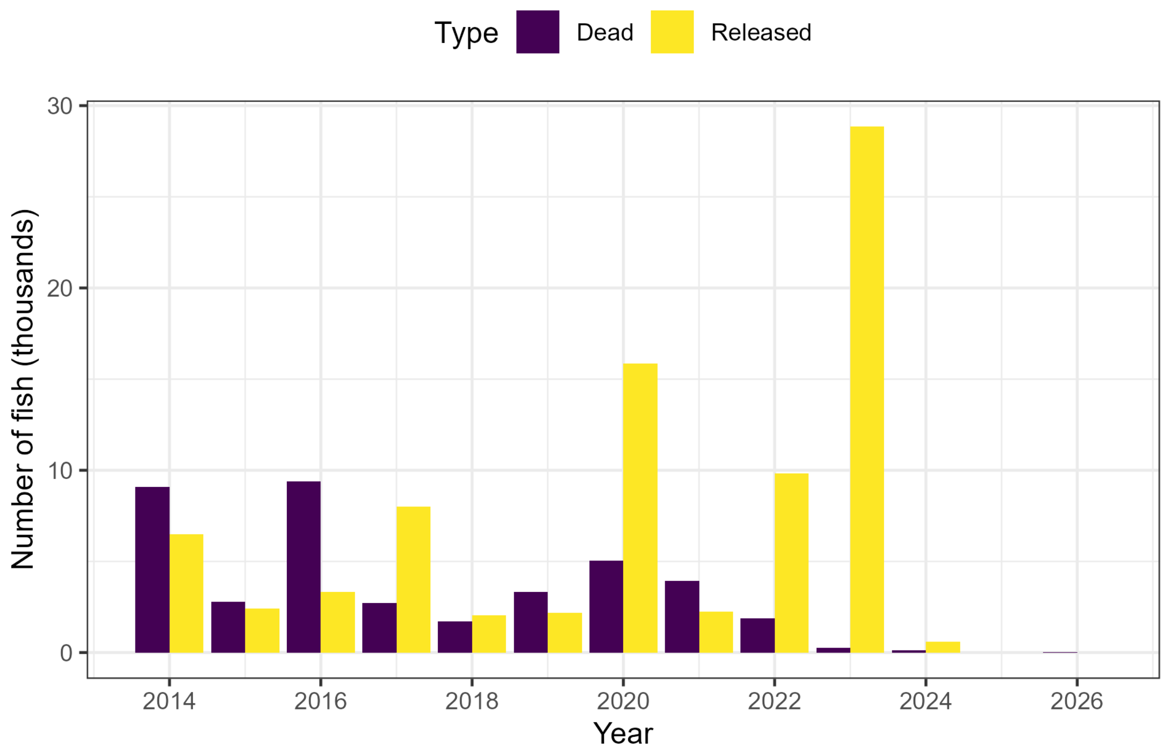

Figure 4. Incidental Pacific Herring mortality in aquaculture activities in thousands of fish from 2014 to 2026 in the Strait of Georgia major stock assessment region (SAR). Note: figure may include data outside SAR boundaries.

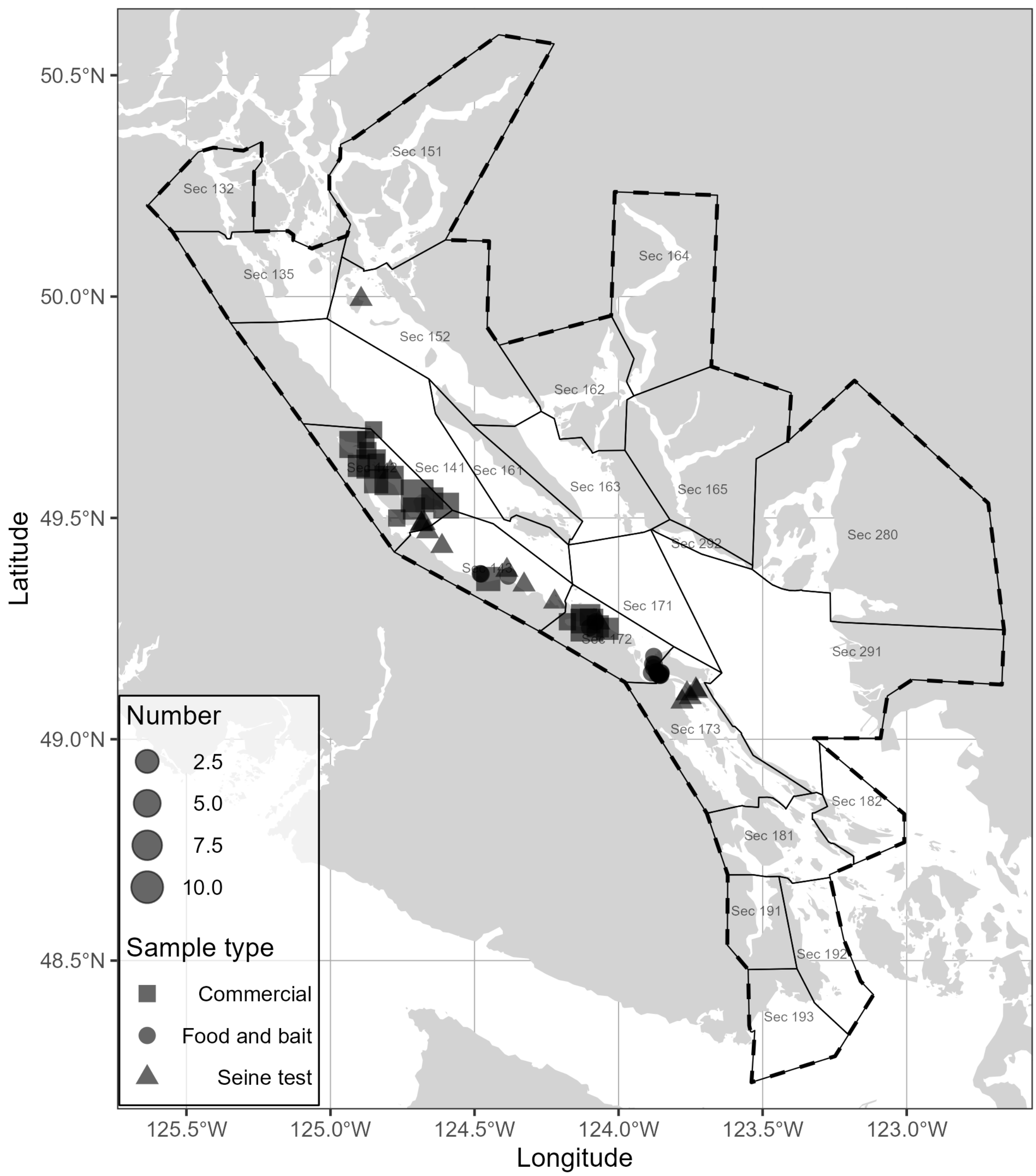

Figure 5. Location and type of Pacific Herring biological samples collected in 2026 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). Units: kilometres (km).

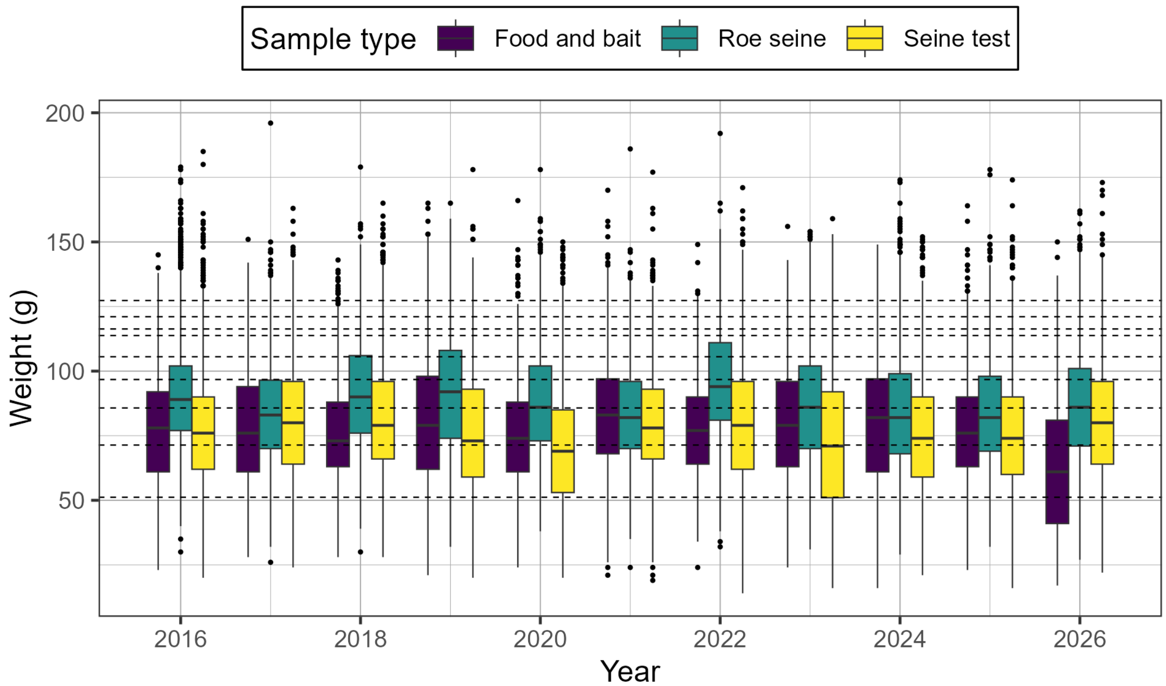

Figure 6. Time series of weight in grams (g) of Pacific Herring by sample type from 2016 to 2026 in the Strait of Georgia major stock assessment region (SAR) in Statistical Areas 14 and 17. The outer edges of the boxes indicate the 25th and 75th percentiles, and the middle lines indicate the 50th percentiles (i.e., medians). The whiskers extend to 1.5 $\times$ IQR, where IQR is the distance between the 25th and 75th percentiles, and dots indicate outliers. Horizontal dashed lines indicate the mean weight-at-age for age-2 (lowest line) to age-10 (incrementing higher from age-2) fish. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a ‘plus group’ which includes fish ages 10 and older.

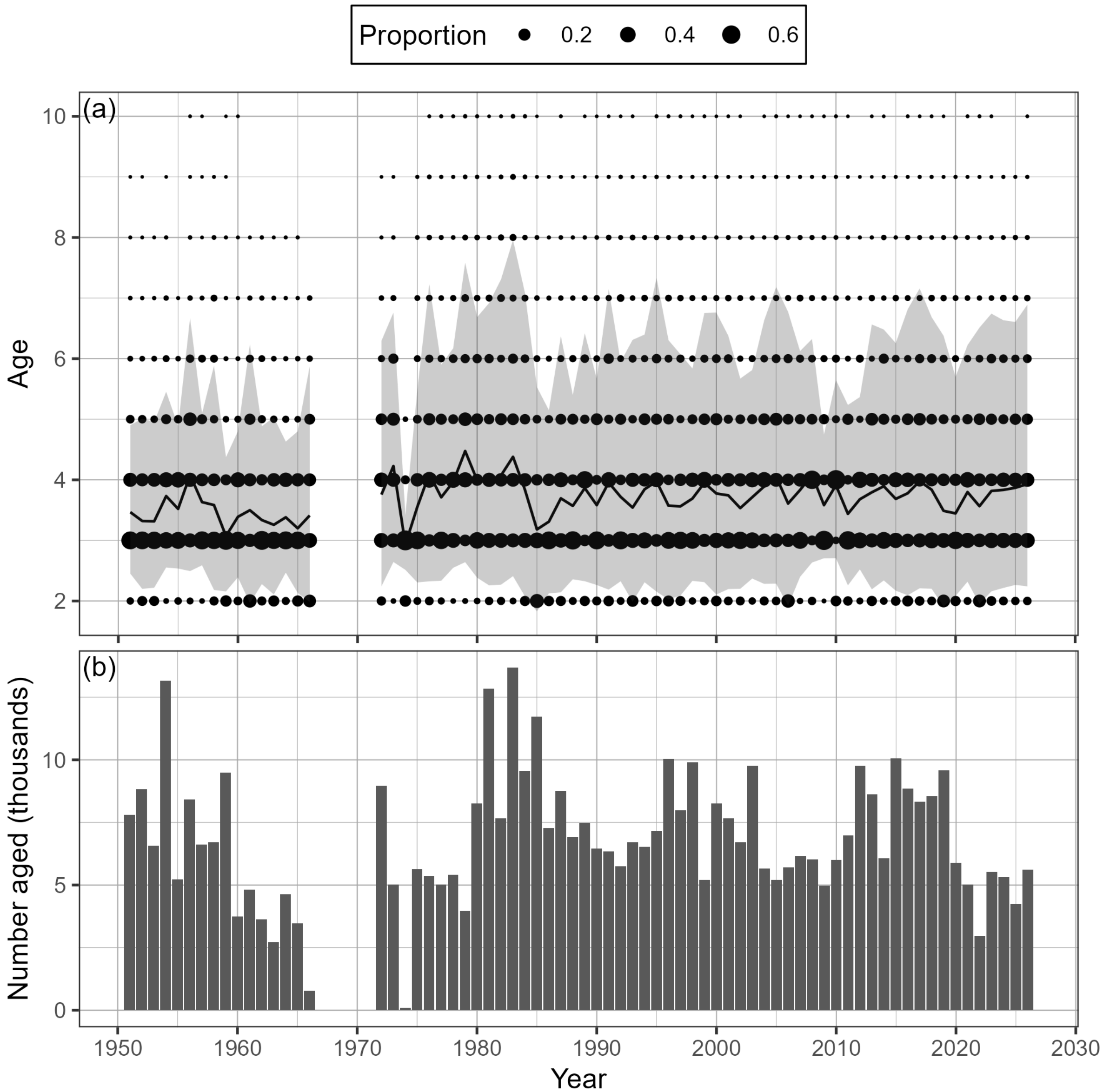

Figure 7. Time series of observed proportion-at-age (a) and number aged in thousands (b) of Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). The black line is the mean age, and the shaded area is the approximate 90% distribution. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a 'plus group' which includes fish ages 10 and older.

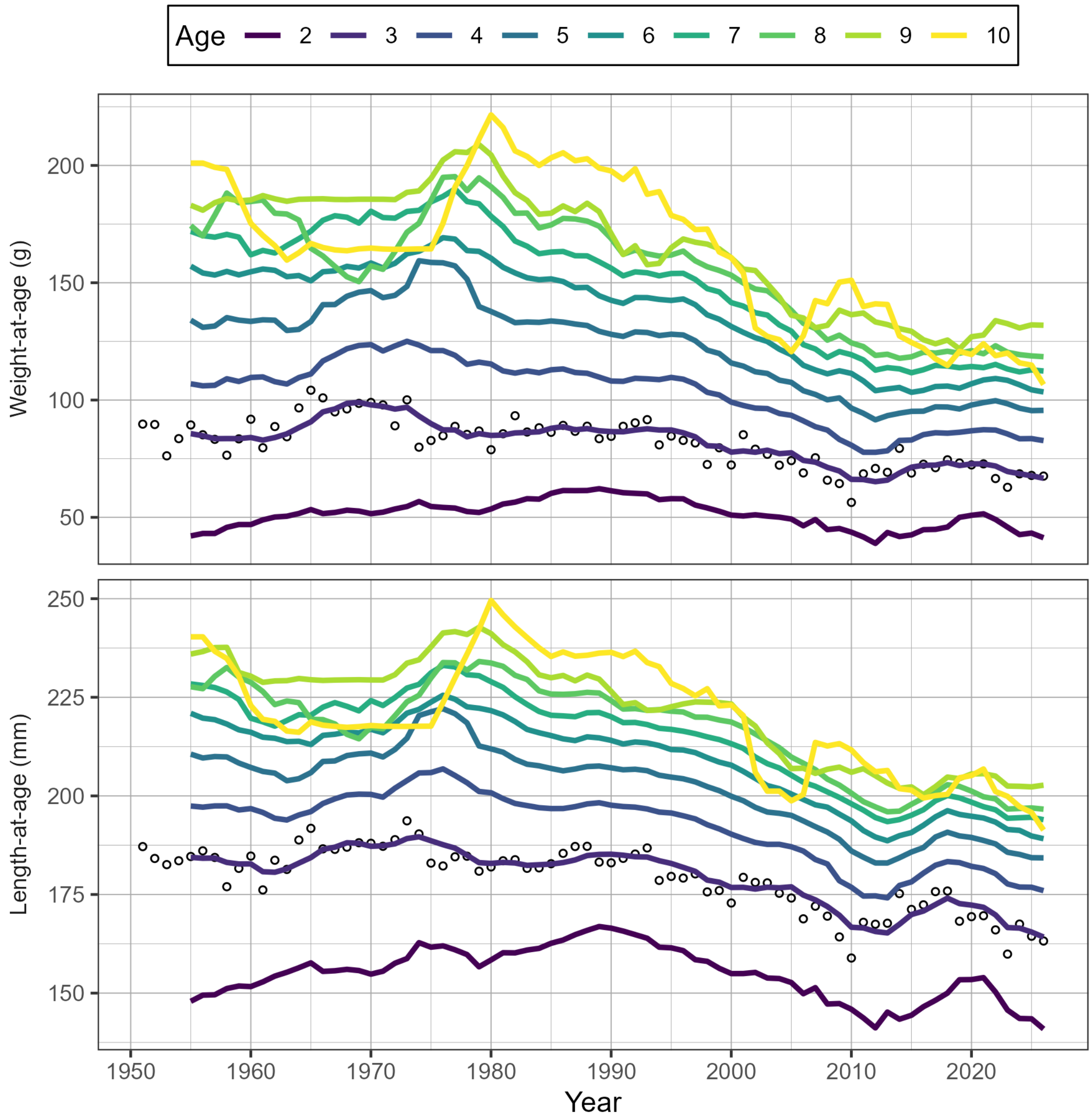

Figure 8. Time series of weight-at-age in grams (g) and length-at-age in milimetres (mm) for age-3 (circles) and 5-year running mean weight- and length-at-age (lines) for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). Missing weight- and length-at-age values (i.e., years with no biological samples) are imputed using one of two methods: missing values at the beginning of the time series are imputed by extending the first non-missing value backwards; other missing values are imputed as the mean of the previous 5 years. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet. The age-10 class is a 'plus group' which includes fish ages 10 and older.

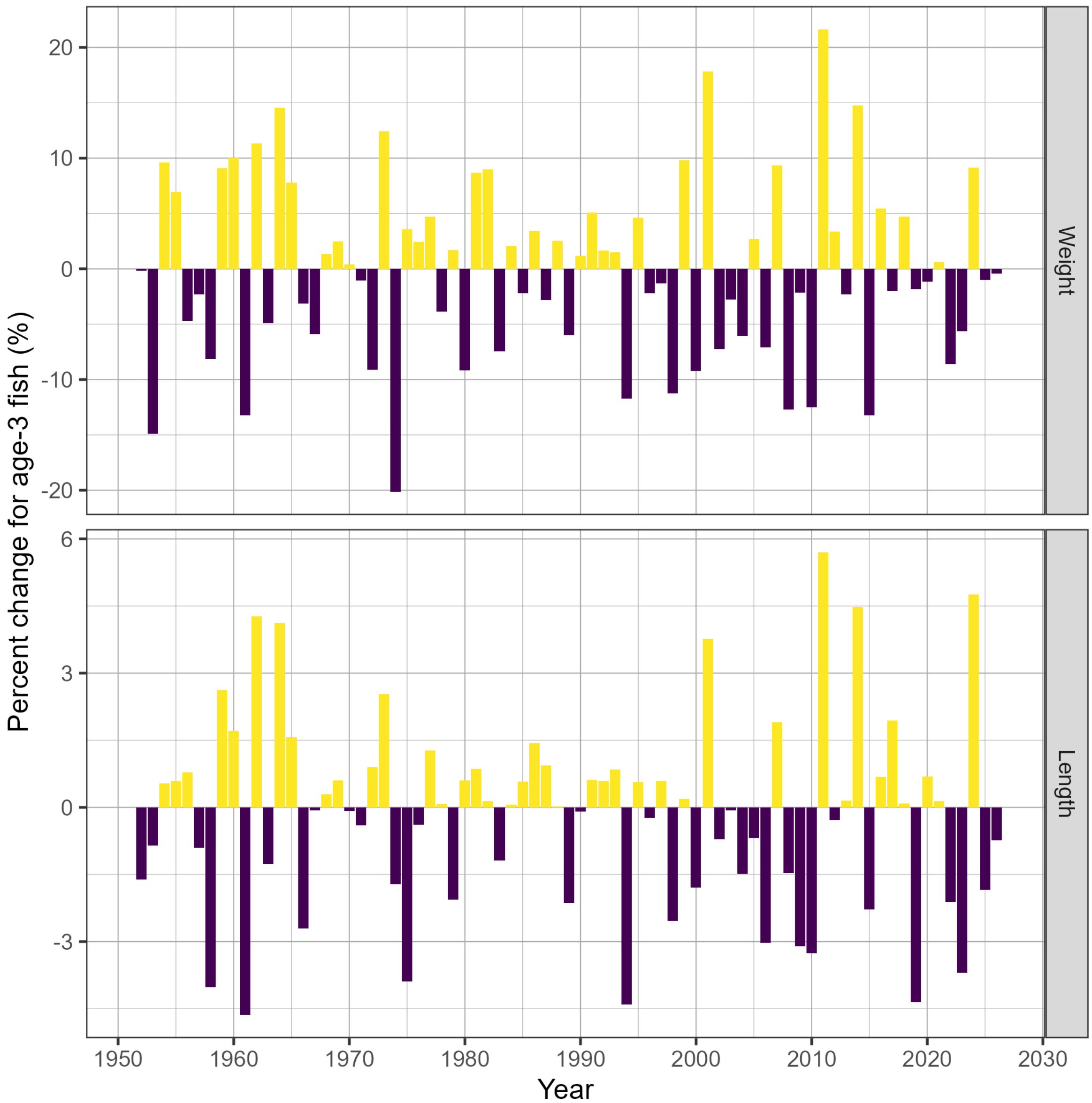

Figure 9. Time series of percent change (%) in weight and length for age-3 fish for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-3 fish, respectively, in year t. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

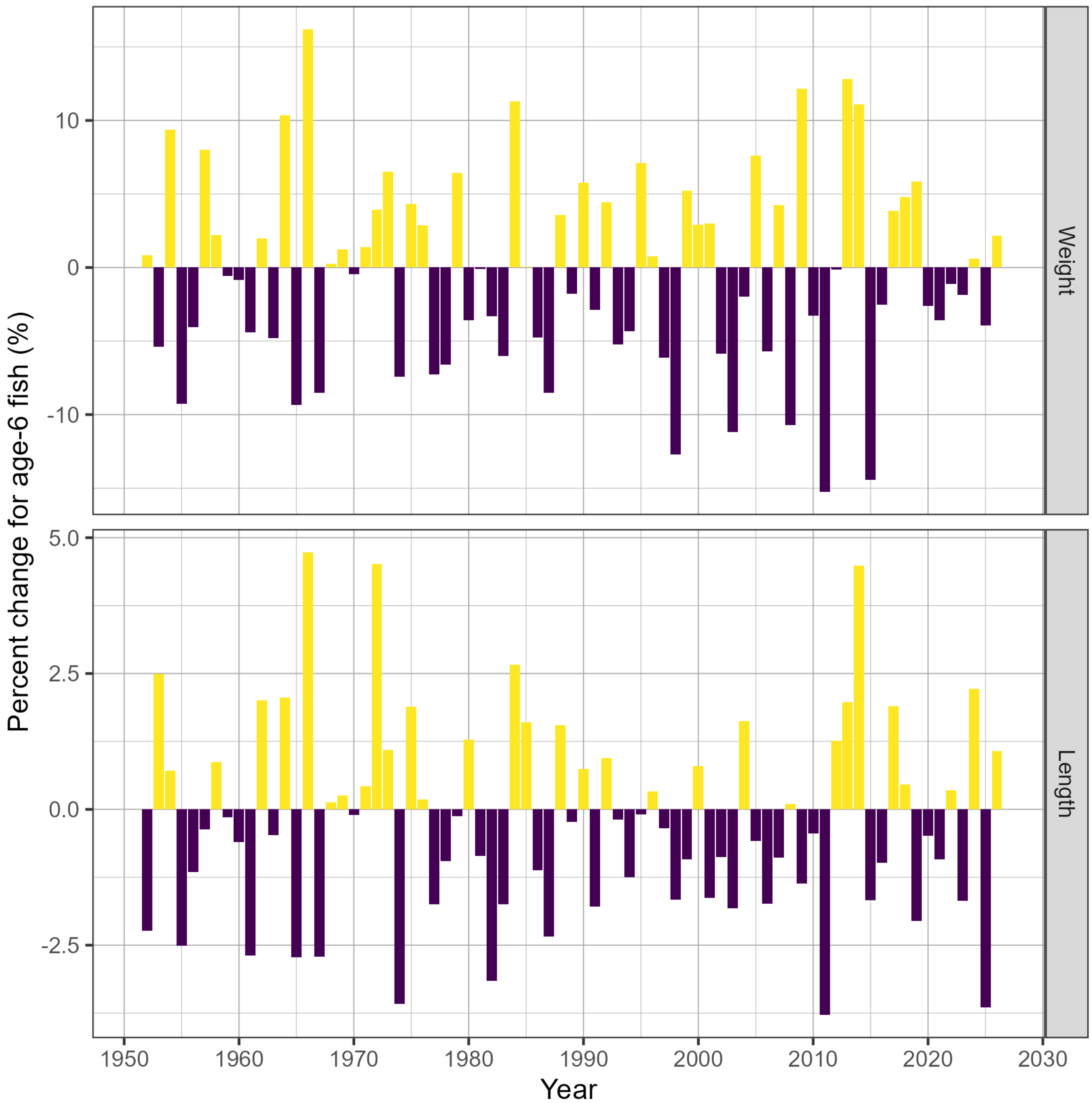

Figure 10. Time series of percent change (%) in weight and length for age-6 fish for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the weight and length of age-6 fish, respectively, in year t. Biological summaries only include samples collected using seine nets (commercial and test) due to size-selectivity of other gear types such as gillnet.

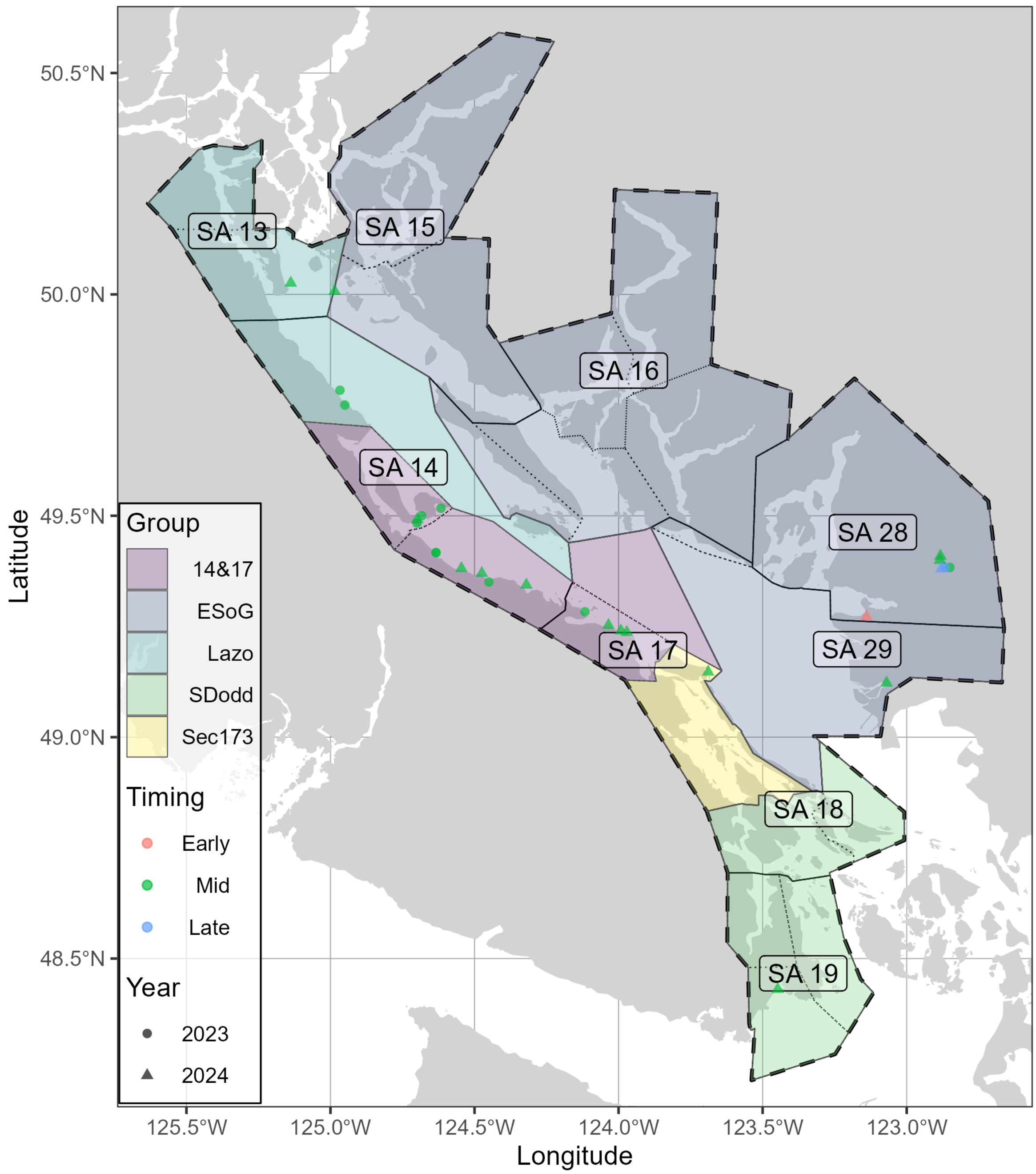

Figure 11. Nearshore genetics sampling in the Strait of Georgia major SAR. Note that timing indicates the collection date: 'Early' is January and February, 'Mid' is March and April, and 'Late' is May, June, and July. In addition, note that genetic samples from 2025 are not yet available. See Figure 2 for legend.

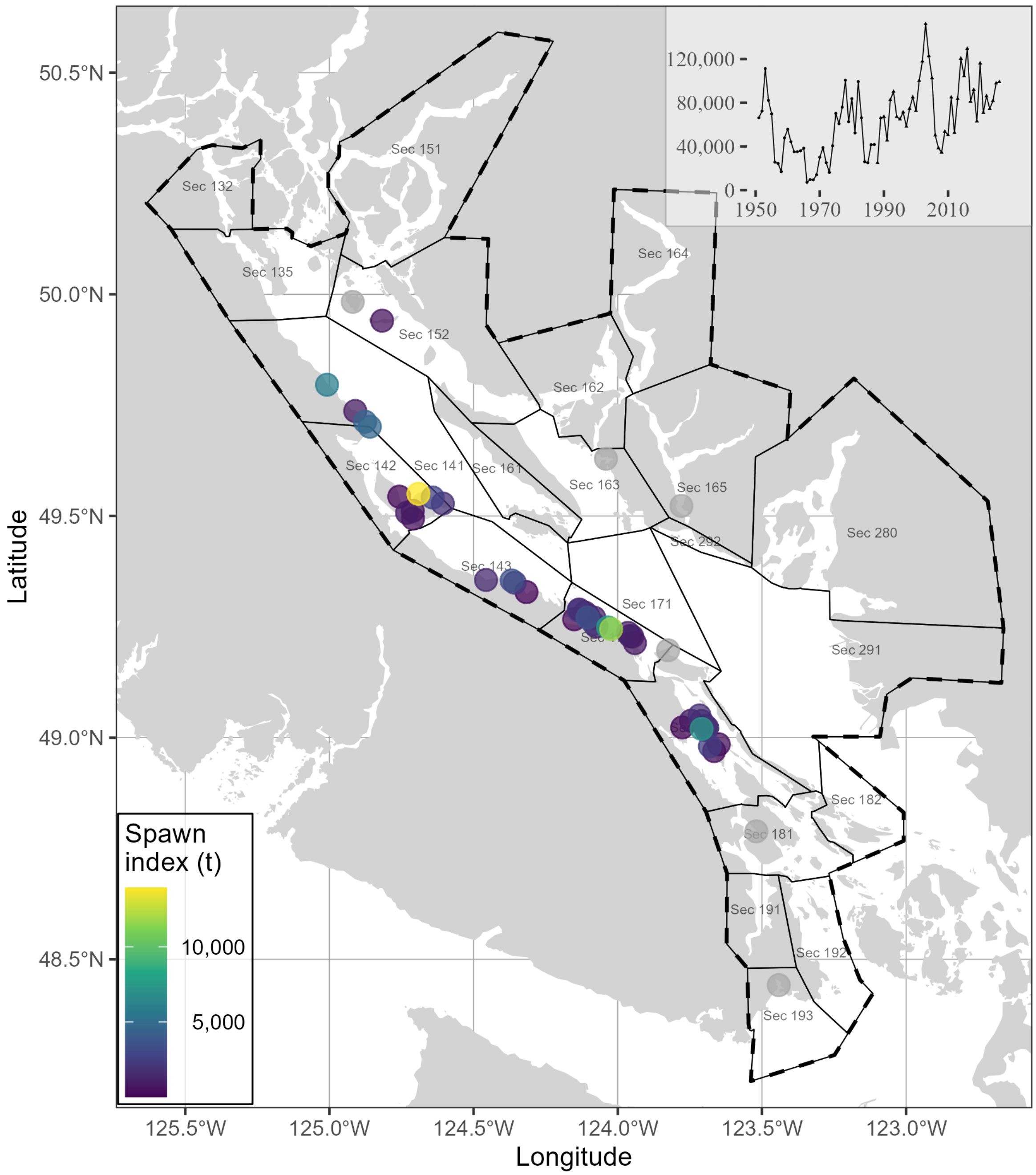

Figure 12. Pacific Herring spawn survey locations, and spawn index in metric tonnes (t) in 2026 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km).

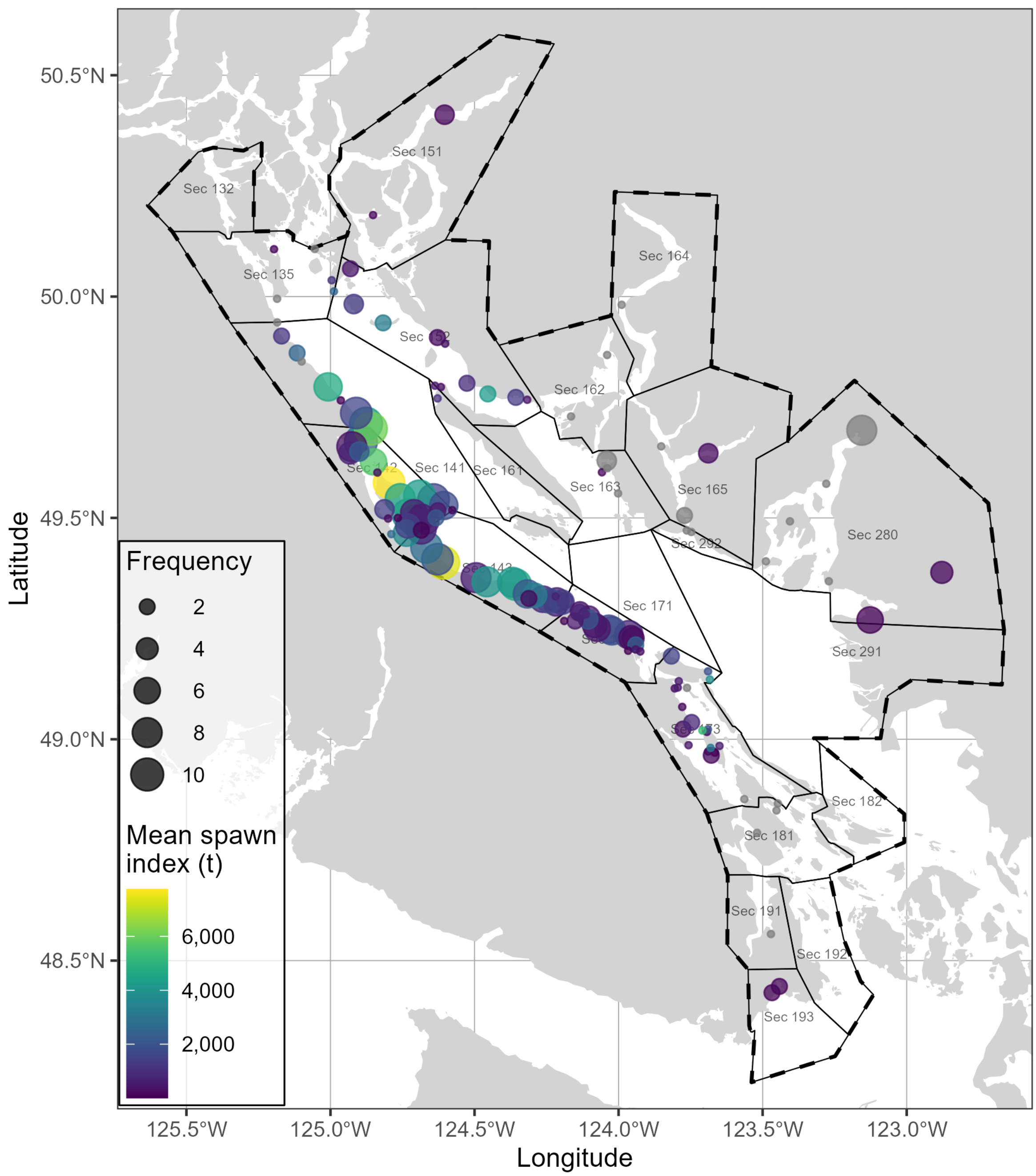

Figure 13. Pacific Herring spawn survey locations, mean spawn index in metric tonnes (t), and spawn frequency from 2016 to 2025 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Units: kilometres (km).

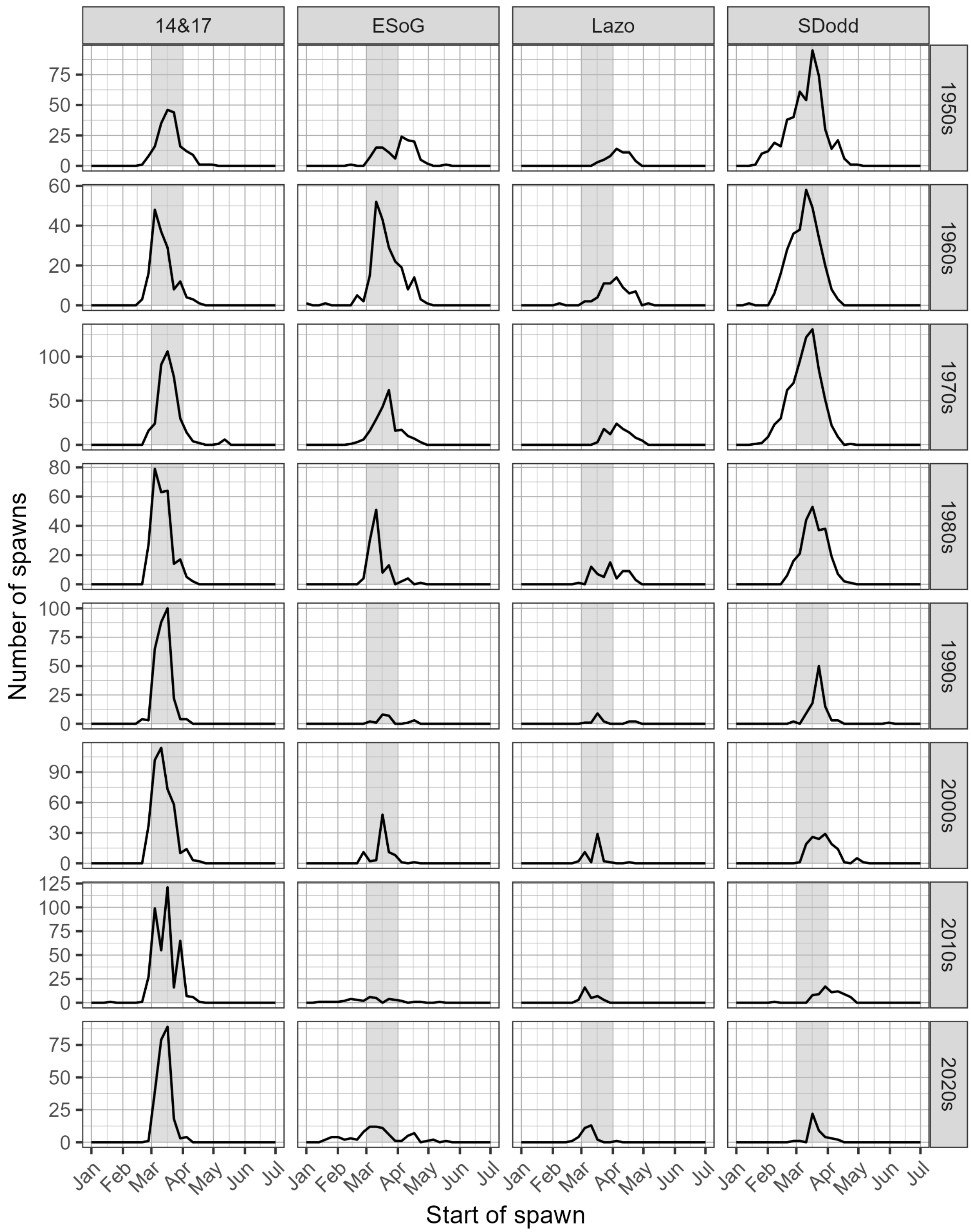

Figure 14. Pacific Herring spawn start date by decade and Group. Grey shaded regions indicate March 1st to 31st. Note that spawn size and intensity varies; therefore the number of spawns is not directly proportional to spawn extent or biomass. Legend: '14&17' is Statistical Areas 14 and 17 (excluding Section 173); 'ESoG' is eastern Strait of Georgia; 'Lazo' is above Cape Lazo; and 'SDodd' is South of Dodd Narrows

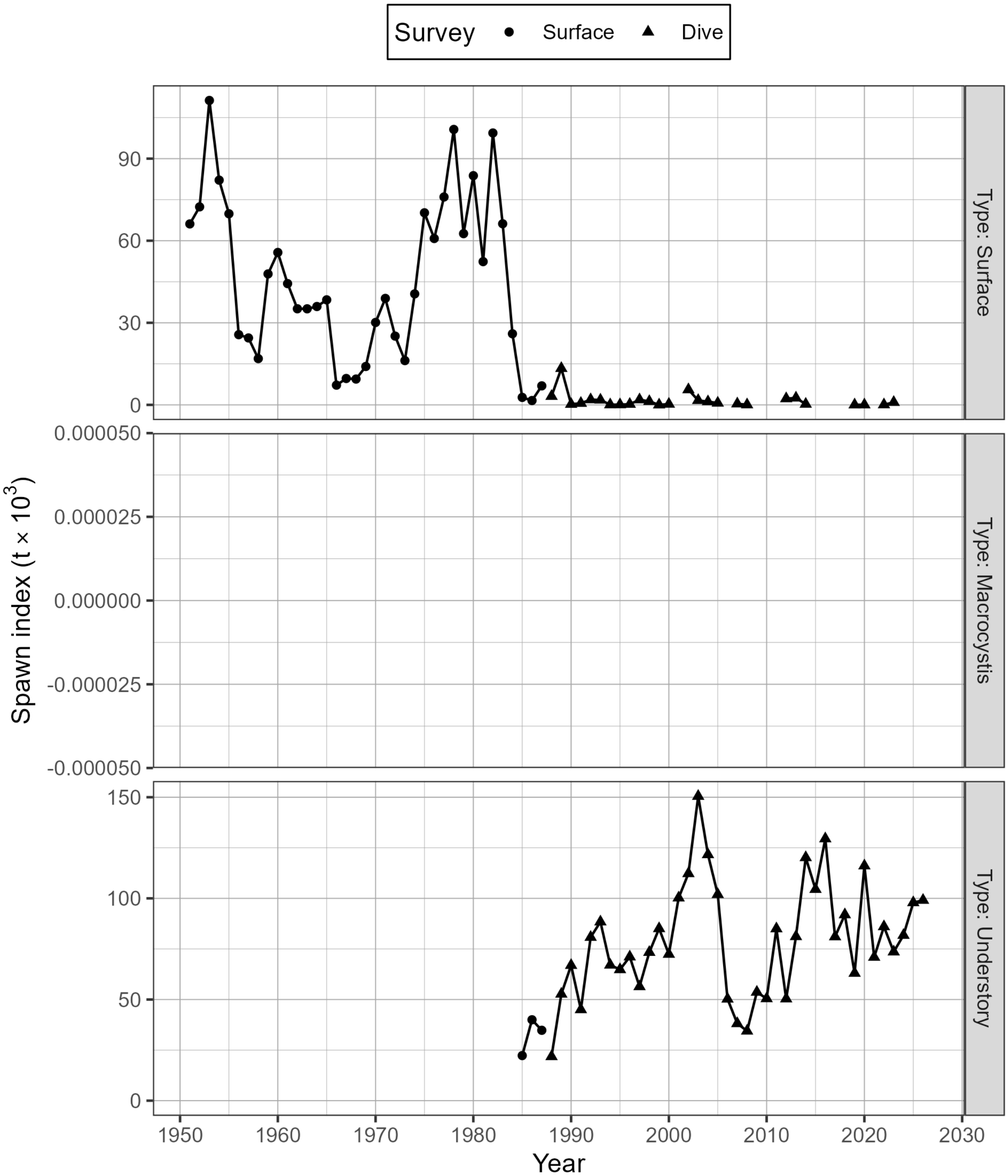

Figure 15. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) by type for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). There are three types of spawn survey observations: observations of spawn taken from the surface usually at low tide, underwater observations of spawn on giant kelp, Macrocystis (*Macrocystis* spp.), and underwater observations of spawn on other types of algae and the substrate, which we refer to as 'understory.' The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). See Figure 16 for the spawn survey method.

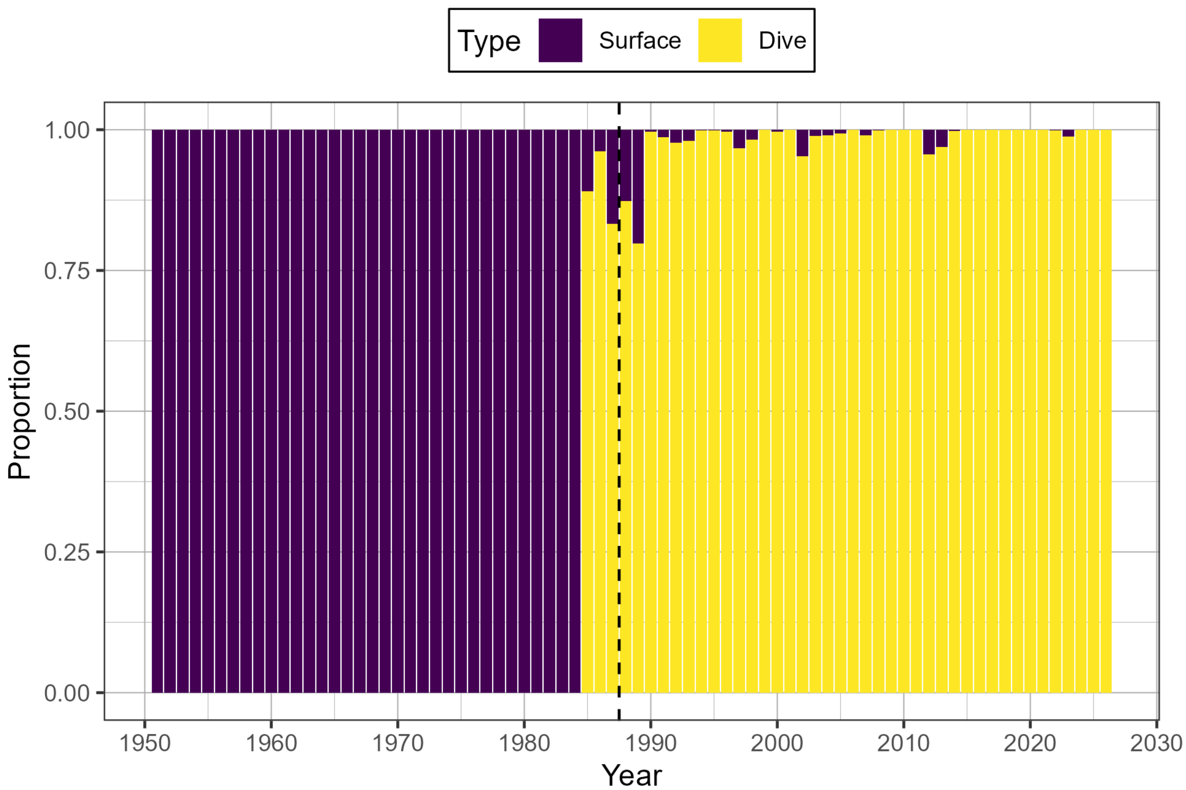

Figure 16. Time series of proportion of spawn index by surface and dive surveys for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

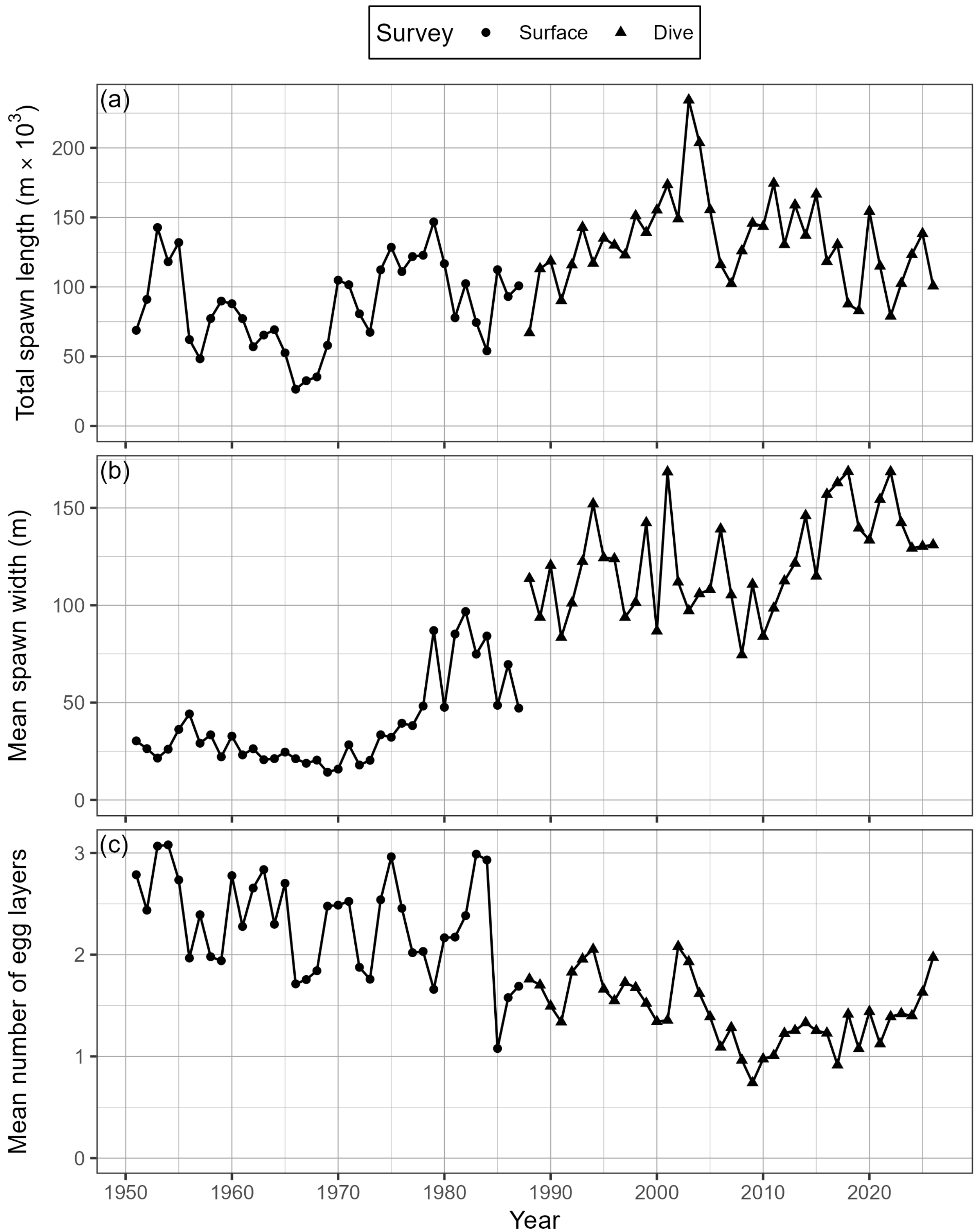

Figure 17. Time series of total spawn length in thousands of metres (m $\times$ 10^3; panel a), mean spawn width in metres (b), and mean number of egg layers (c) for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026).

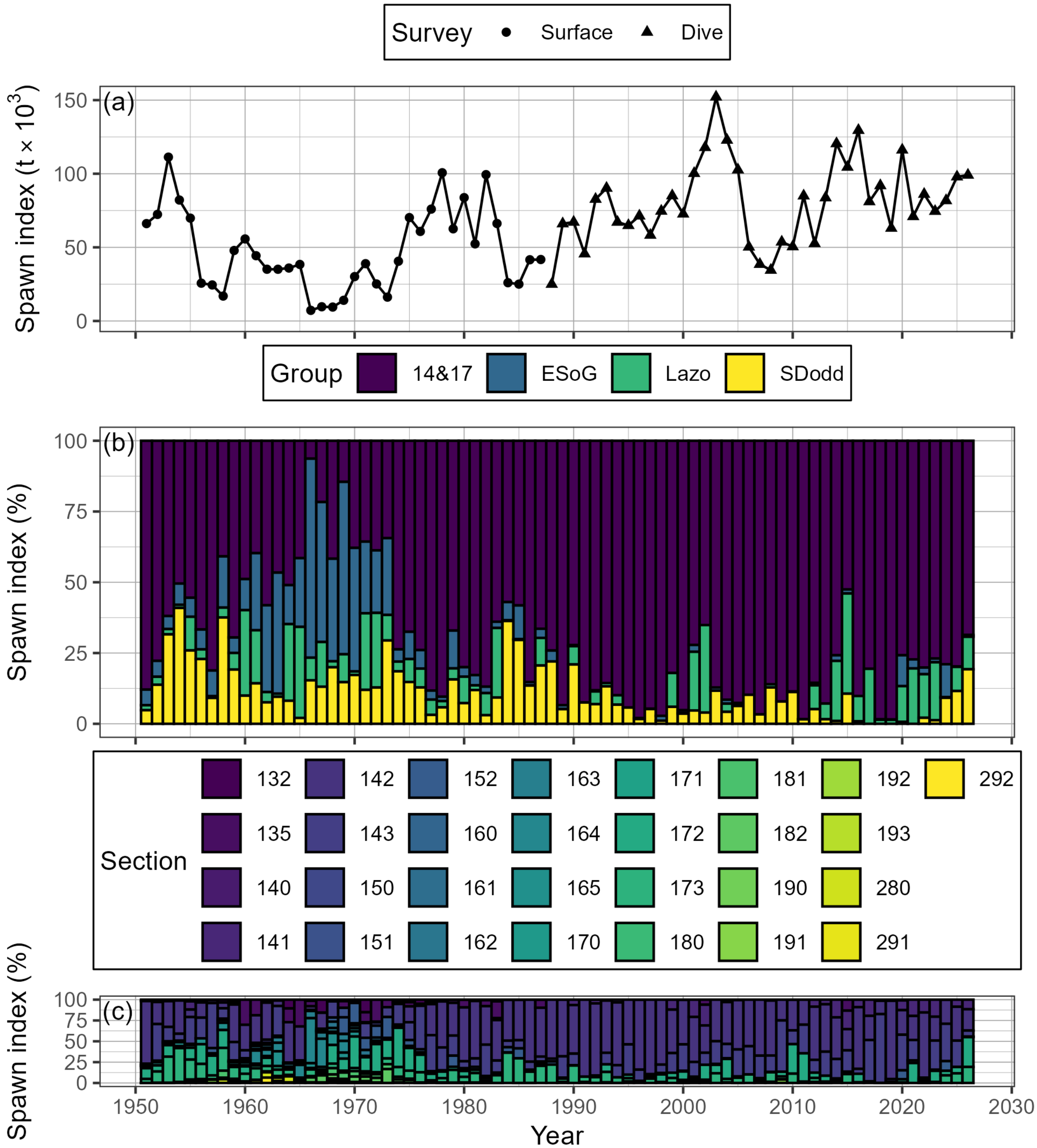

Figure 18. Time series of spawn index in thousands of metric tonnes ($t \times 10^3$) for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR; panel a), as well as percent contributed by Group, and Section (b, & c, respectively). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 15 and 16). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Legend: '14&17' is Statistical Areas 14 and 17 (excluding Section 173); 'ESoG' is eastern Strait of Georgia; 'Lazo' is above Cape Lazo; and 'SDodd' is South of Dodd Narrows

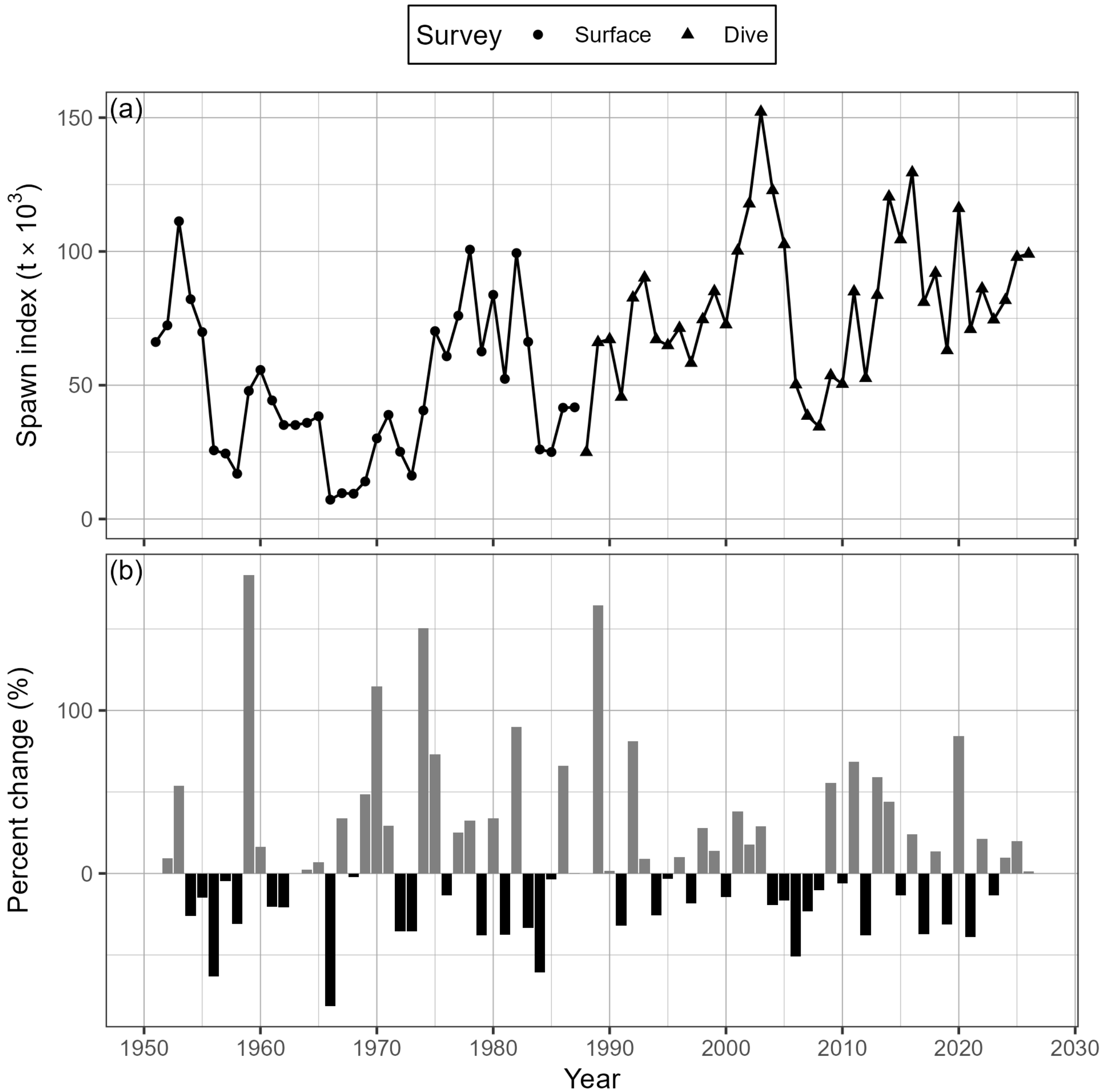

Figure 19. Time series of spawn index in thousands of metric tonnes (t $\times$ 10^3) for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR; panel a), and percent change (b). Percent change is $\delta_t = \frac{\alpha_t - \alpha_{t-1}}{\alpha_{t-1}}$ where α_t is the spawn index in year t. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Note that spawn surveys in the dive survey period (1988 to 2026) are a combination of surface and dive surveys (Figures 15 and 16).

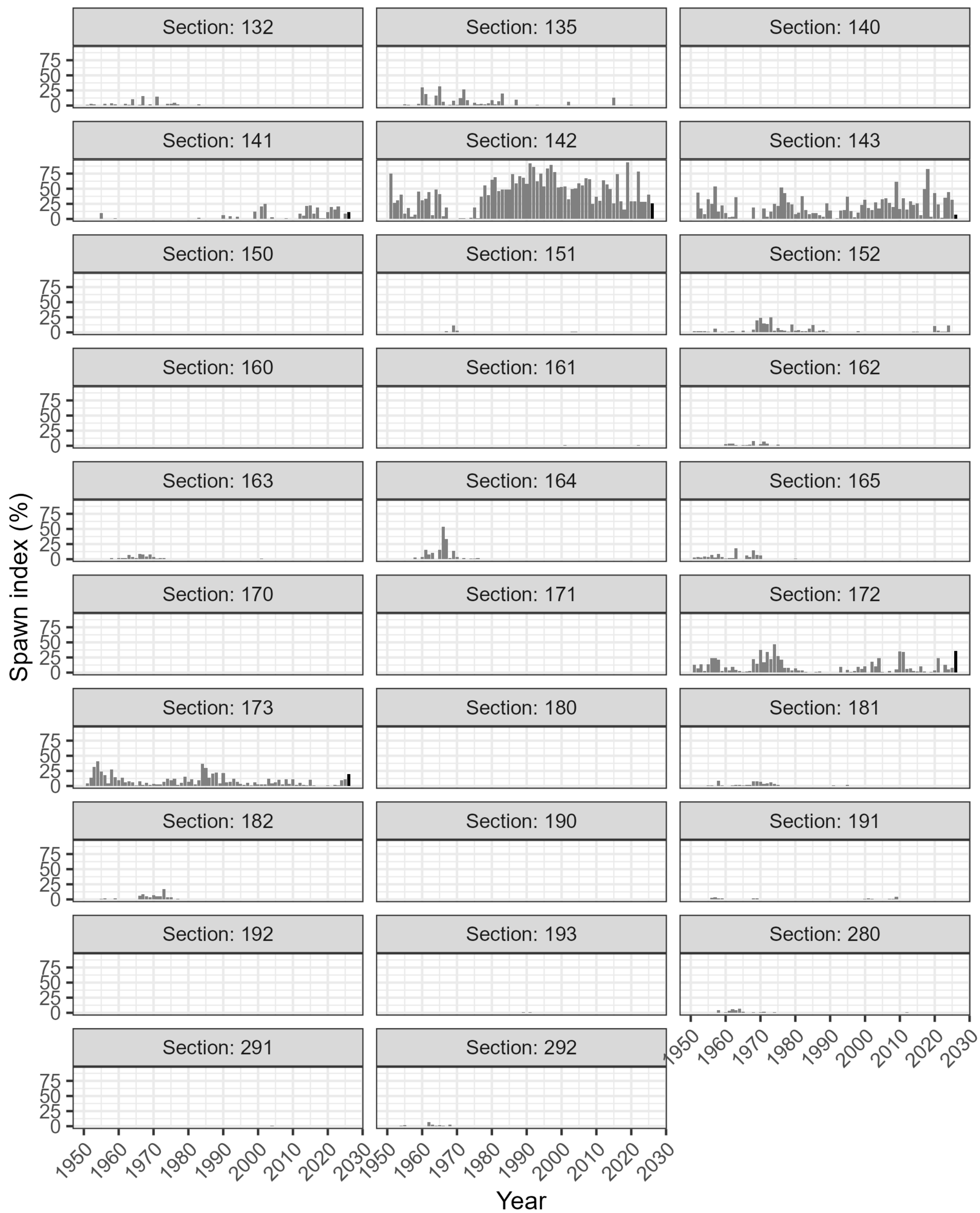

Figure 20. Time series of percent of spawn index by Section for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). The year 2026 has a darker bar to facilitate interpretation. The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q.

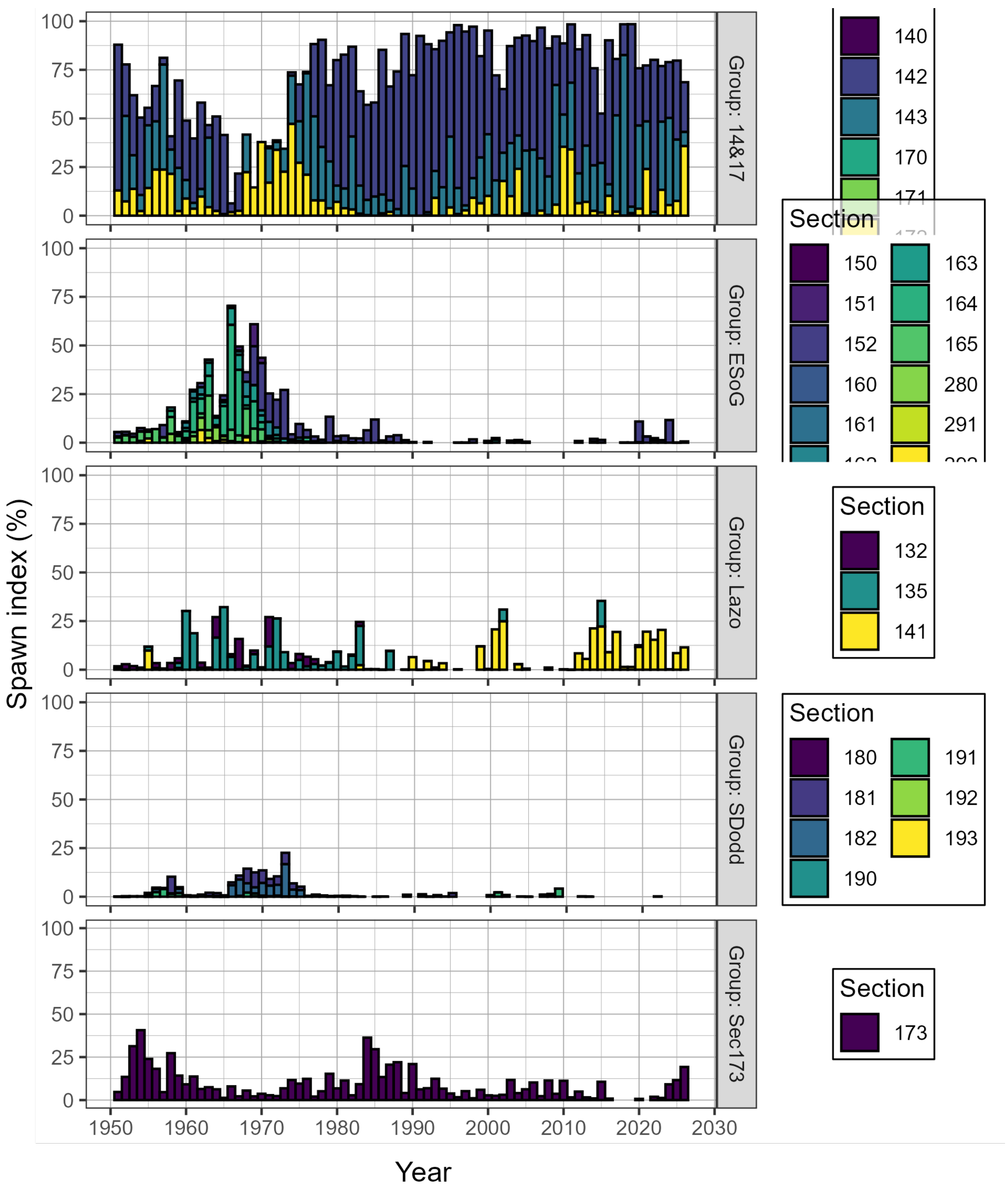

Figure 21. Time series of percent of spawn index by Group and Section for Pacific Herring from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Legend: '14&17' is Statistical Areas 14 and 17 (excluding Section 173); 'ESoG' is eastern Strait of Georgia; 'Lazo' is above Cape Lazo; and 'SDodd' is South of Dodd Narrows

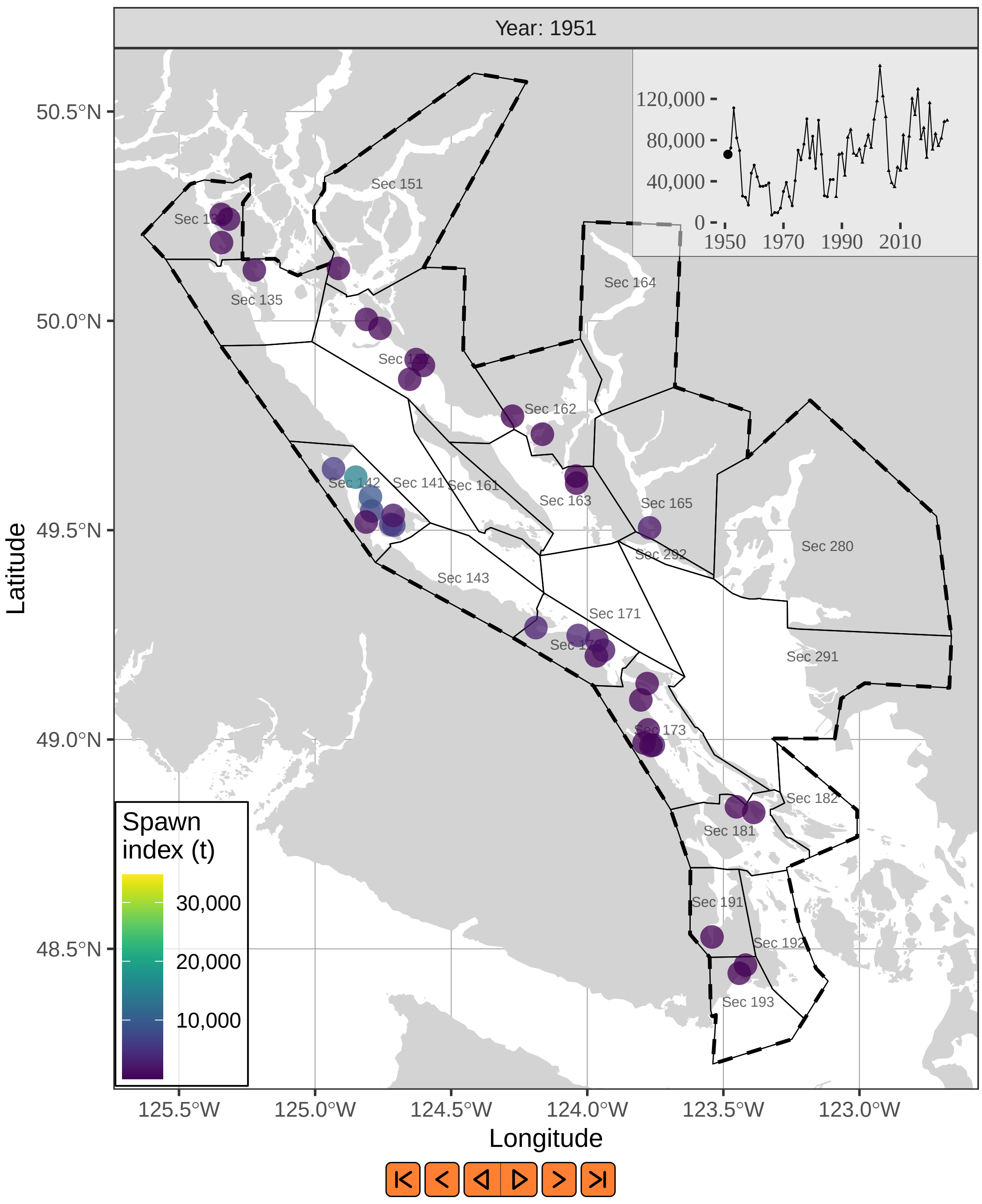

Figure 22. Animation of Pacific Herring spawn index in metric tonnes (t) by Location from 1951 to 2026 in the Strait of Georgia major stock assessment region (SAR; thick dashed lines), and associated Sections (Sec; thin solid lines). The spawn index has two distinct periods defined by the dominant coastwide survey method: surface surveys (1951 to 1987), and dive surveys (1988 to 2026). The 'spawn index' is not scaled by the spawn survey scaling parameter, q. Missing spawn index values indicate incomplete surveys (grey circles). Inset tracks the total spawn index. Units: kilometres (km). View the animation: download the report, open with Adobe, enable Java, and click "play".